

Relaxing the steady-state assumption displaces forest soil carbon models along the trade-off between litter input and turnover

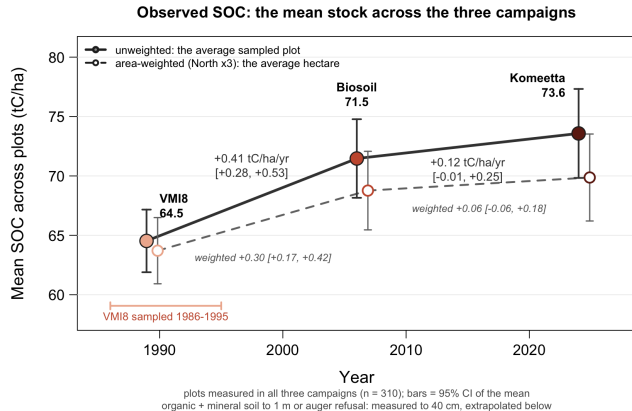
HIKET — internal progress talk: where the manuscript stands

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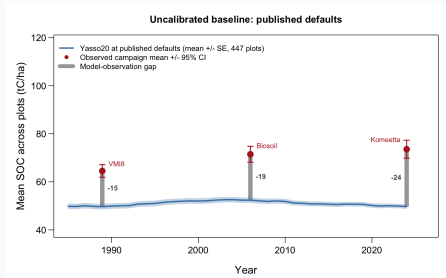
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Finnish forest soils have been measured three times



Every plot equal (solid, compared with the models): 64.5 → 71.5 → 73.6 tC/ha. Weighted by forest area (dashed): 63.7 → 68.8 → 69.9. Either way the second interval gained much less than the first: the accumulation is slowing down.

The convention: start the soil in balance with its litter input

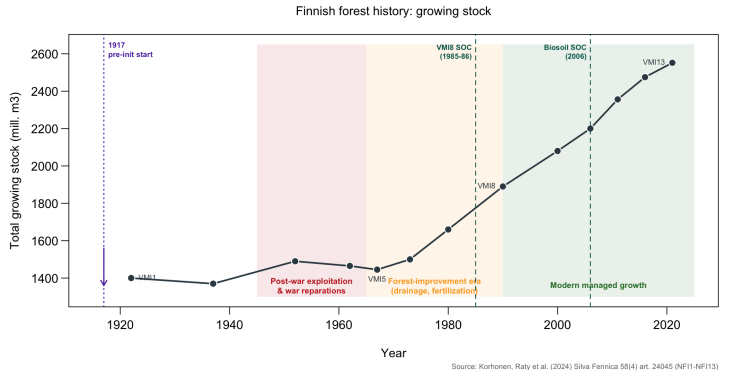


Yasso20, published parameters, steady-state start: flat and low.

- Shared across models: Yasso, RothC, Century, global land-model spin-ups; CBM-CFS3 cycles to a repeating average.
- The Finnish inventory spins up on NFI6 litter (1971–76) and 1960–90 climate; its defence is Peltoniemi et al. 2006 — a **precision** study around an assumed level (“precision, not accuracy”).
- Liski et al. 2006 is our direct antecedent: same country, model family and NFI litter — and they *assumed* the 1922 state, having no repeated soil measurement.

A soil started in balance has no accumulation left to perform.

Finland is the wrong place for that assumption



Growing stock +84% over a century, most of it after the late 1960s. Litter follows biomass, so the input the models must integrate has been rising the whole time — and unmanaged stands hold ~40% more mineral-soil C (Kumpu et al. 2025).

Two changes to the starting state, three questions

What we change

1. The starting state is **estimated**, not assumed: the input the soil is in balance with is a free parameter, informed by the measured trajectory.
2. The date of that balance is **pushed back to 1917**: 68 years of simulated history before the first measurement for any error in it to fade.

Two quantities govern a stock: litter input J and mean transit time τ , $C^* = J\tau$ at balance. A stock fixes the product only; a trajectory can do more.

- I Where does the measured trajectory place the models on the **trade-off between litter input and transit time**, relative to their published parameterisations — and how far does it narrow it?
- II Does that position matter **beyond the observed window**: do two calibrations that fit equally well extrapolate differently, and for which structures?
- III What part of the variation between **individual plots** is left unexplained, and what information would a model need to explain it?

Six models, two roles, one frame

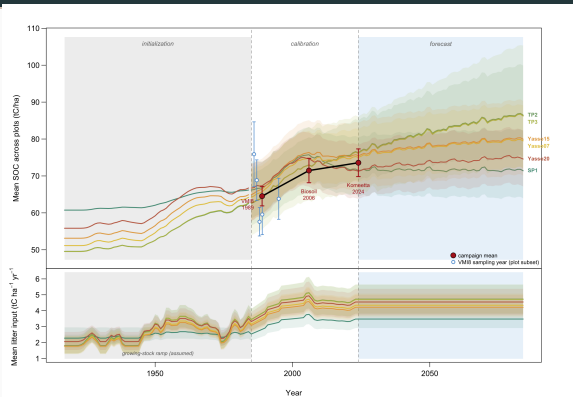
Model	Pools	Free	Role
SP1	1	6	benchmark
TP2	2	7	benchmark
TP3	3	9	benchmark
Yasso07	5	20	the inventory model
Yasso15	5	26	successor
Yasso20	5	26	successor

The simple models are a transparent ladder: any Yasso result is checked against a structure one can follow by hand.

As comparable as the models allow

- Same information content is impossible; we reduce the asymmetry with compromises.
- Yasso: rates **fixed** at published values as part of the model definition; 12 fractions, climate, woody size free.
- SP1/TP2/TP3: ICBM rates — fast **fixed**, slow **tightly pinned** (arable → forest); humification fractions free.
- Roughly same prior rule per class (centres from each model's own publication).
- All six solve the linear system exactly.

The main point: estimate the starting state instead of assuming it



Start in 1917 in balance with the 1917 litter; raise the input along the Liski et al. 2006 reconstructed shape to 1985; then drive with the annual record. Two auxiliary parameters: $\sigma_{\text{init}} = J_{1917}/J_{1985}$ and σ_{input} , a multiplier on the whole litter series.

Observations

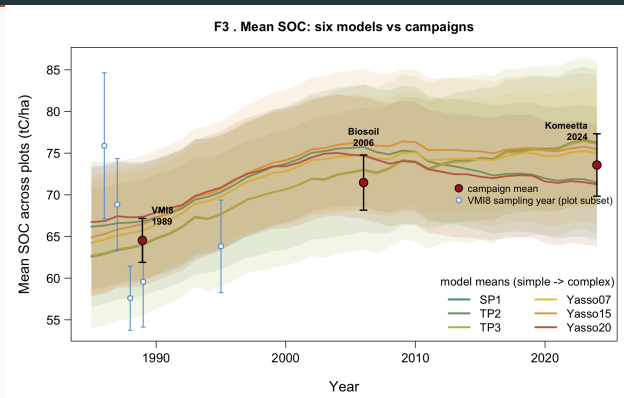
- 456 NFI plots, 1205 plot-years; 365 fitted / 91 held out (by plot).
- Three campaigns rebuilt on **one basis**: bulk density, coarse fragments (median 43%, a $1.6\times$ effect), per-plot depth, modelled deep tail (validated on held-out 40–80 cm).
- Official 2006/2024 national stocks reproduced to 0.01 t/ha. First campaign: 2006 litter+moss layer added; **true sampling year** used.

Litter and climate

- Tupek et al. tree-litter product, AWEN $\times$ size class; $\bar{J} = 2.511 \text{ tC ha}^{-1} \text{ yr}^{-1}$; understorey *not* included by design.
- FMI 10 km daily grid; climate modifier computed per year, then averaged (Jensen).

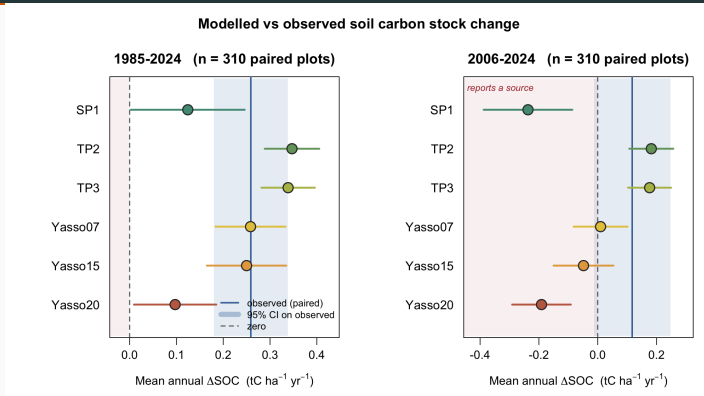
Error model: observations sharing a plot, a region or a campaign share part of their error (backup).

All six models reach the observed stocks



No model sits more than 3.5 tC/ha from any campaign mean — with anchored rates, a constrained input and an estimated start. A precondition, not a result: any $J \times \tau$ with the same product reaches the same stock.

... and bracket the observed rate of change



Observed +0.259; Yasso07 +0.258, Yasso15 +0.250; TP2/TP3 stronger, SP1/Yasso20 weaker. The bracketing is the result. Which model lands closest is *not* interpretable — reported as consistency, never validation.

Model structure is diluted by local variability

Model	Pools	Free	R^2 cal	R^2 hold	RMSE	Eff. flux
SP1	1	6	0.017	0.000	47	3.2
TP2	2	7	0.016	0.000	48	4.3
TP3	3	9	0.015	0.000	49	4.3
Yasso07	5	20	0.019	0.000	48	4.0
Yasso15	5	26	0.021	0.001	47	3.9
Yasso20	5	26	0.016	0.000	47	4.2

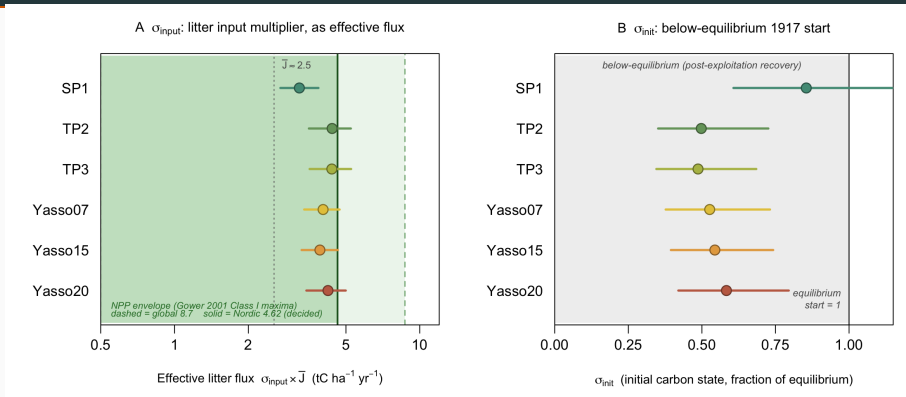
Table 1 of the draft. Effective flux = $\sigma_{\text{input}} \bar{J}$, $\text{tC ha}^{-1} \text{yr}^{-1}$; RMSE on held-out plots, tC/ha .

- Held-out R^2 is zero to three decimals in all six; RMSE spans 3.6% best to worst.
- Skill does not climb the pool ladder. The fit cannot choose a structure.
- Fractions converge ($R\text{-hat} \leq 1.008$) yet stay prior-pinned, anti-correlated within a pool, < 1 nat gained.

What three campaigns identify

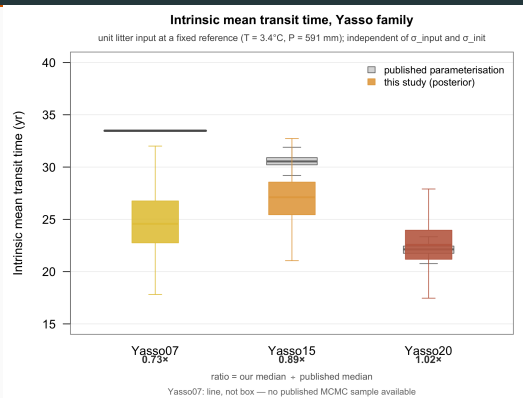
a stock and a rate of change — which is what the inventory needs, and all that is claimed.

What reproducing the trajectory requires: more litter input...



Left: effective flux 3.2–4.3 tC ha⁻¹ yr⁻¹ in all six, against the 2.5 the product supplies — inside the NPP envelope. Right: σ_{init} 0.49–0.60 (SP1 0.83) against ~ 0.90 from the growing-stock record — read as an **effective starting deficit** of the 1917 soil.

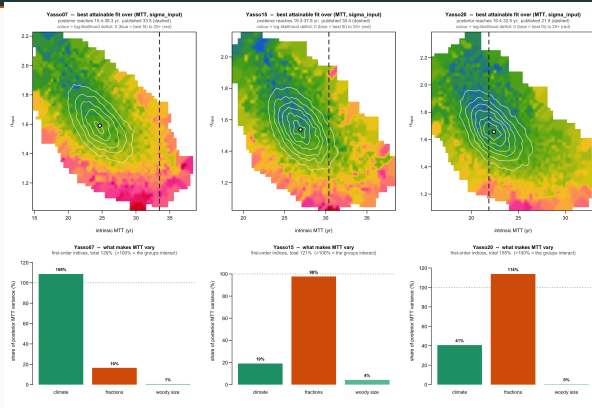
... and, for two of three Yassos, faster turnover



Intrinsic mean transit time at one reference climate and litter mix, from the equations alone.

Calibrated / published: 0.73 (Yasso07), 0.89 (Yasso15), 1.03 (Yasso20). **Yasso20 is the negative control** — a data-treatment artefact would have moved all three.

The ridge we are moving on



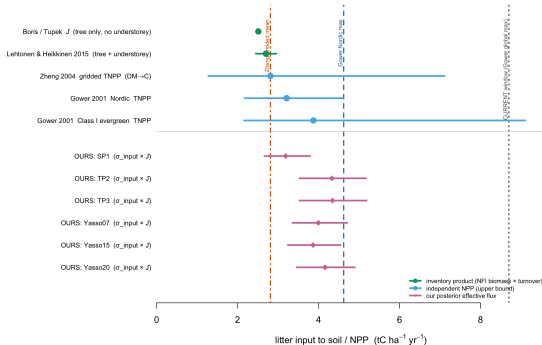
Colours = fit to the data alone; white contours = posterior (data + priors); dashed = published MTT. Going back to the published MTT costs little: ≤ 9.7 / ≤ 3.0 / 0 log-likelihood units. The data **suggest a direction rather than impose a value.**

What might be missing from the litter input

Litter input to Finnish forest soil: inventory products vs independent NPP vs our calibration

Bars are reported ranges (Gower, Zheng) or 95% intervals (L&H, ours).

$\text{litter} = \text{NPP} - \text{biomass increment} - \text{stemwood removed} - \text{exudates}$.

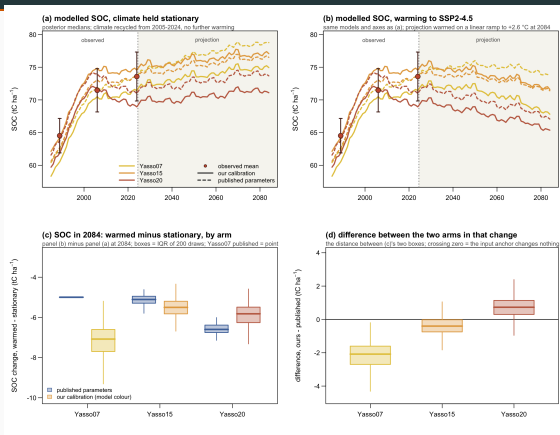


Gap to explain: $0.68\text{--}1.83 \text{ tC ha}^{-1} \text{ yr}^{-1}$

- **Understorey** — omitted by design; 0.51–0.67 in the NID itself.
- **Mycorrhizal residues** — not a biomass compartment; plausibly 0.2–0.5.
- **Rhizodeposition** — standard in cropland accounting (Bolinder 2007); absent from the forest chain.
- **Fine-root lifespan** — points the *other* way ($\sim 1/3$ of the turnover in use).
- **Early succession** — a timing gap after clear-cut.

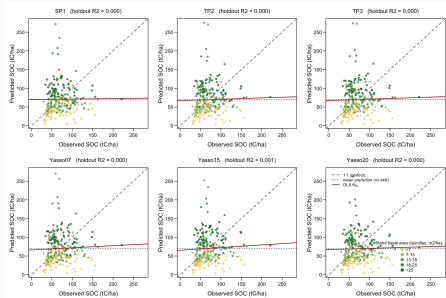
A set that brackets the gap, not a sum that matches it.

Published vs calibrated, run forward under the same warming



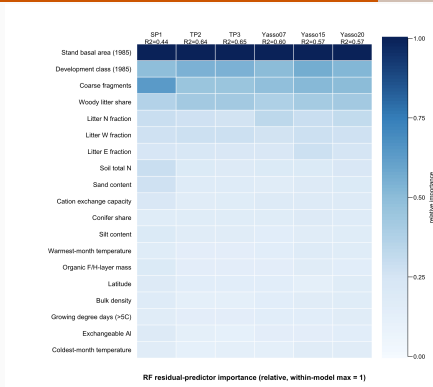
Neither arm recalibrated; only σ_{input} refitted so both reproduce the observed window. Only **Yasso07** separates (-2.1 tC/ha, $[-3.6, -0.8]$, SSP2-4.5). Yasso15/20 do not separate at all.

The models reproduce the national mean, not the differences between plots



- Held-out $R^2 = 0$ in all six. What matters is what the models *predict*: stocks ordered by basal area, spread wider than observed.
- Slope of log stock on basal area: models $+0.044$ – 0.049 , observed $+0.0044$ (n.s. after region and soil).
- No model has a basal-area term: the **litter product** rises at $+0.072$ ($\times 18$ across the range) and the models pass it through.
- Woody-compartment mismatch **refuted**: zeroing woody inputs moves it 6%; SP1/TP2/TP3 have no composition yet give the same slope.
- Plots cut in 1985–95 hold $\sim 40\%$ more C than predicted: the same missing flux, from the other end.

About half of what the models leave behind is recoverable



A random forest on the residuals recovers 44–65% out of bag, against an observation repeatability of ~ 0.49 . The same variables lead in all six: basal area, development class, coarse fragments. A productivity proxy and a soil property no model has a term for.

Conclusions

1. An **estimated** transient starting state lets all six models represent the three campaigns and bracket the observed rate. The fit cannot separate the structures.
2. Reproducing the trajectory places the models at a **larger litter input** and, in two of three Yassos, faster turnover, an equifinality trade-off the soil data cannot resolve. My most probable reading: the litter input is **incomplete** (root exudation, understorey, mycorrhizal residues, early succession).
3. The position on the trade-off **changes the projection of Yasso07 only**: one climate modifier ties turnover to temperature sensitivity; pool-specific modifiers are robust to the choice.
4. Under constant input the **sink is saturating**: soil 26–42% below balance, sink falls by a third to a half in 60 yr. How much remains depends on the estimate of 1917 state. +20% input buys 6–10 tC/ha by 2050 — finite.
5. A **baseline for model development**: half the plot-scale residual is recoverable from existing site variables — stand productivity and soil physical properties.

Where the manuscript stands, and what is still open

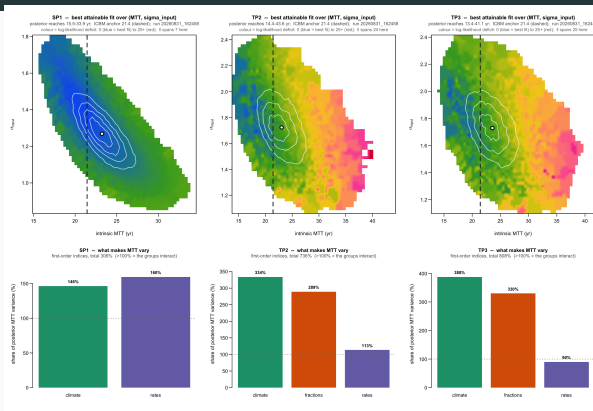
Done

- Full draft, Introduction to Conclusions; first full review (72 notes) addressed.
- Abstract, Limitations and the maps paragraph drafted.
- Title chosen; mean *transit* time adopted (Sierra et al. 2017).
- Literature review, 8 topics; missing-flux budget; climate source confirmed (FMI 10 km grid).
- Forward scenarios (constant and +20% input, equilibrium stock) for all six.
- NextGenC deliverable rebuilt on the production run (maps, matrices).

Open

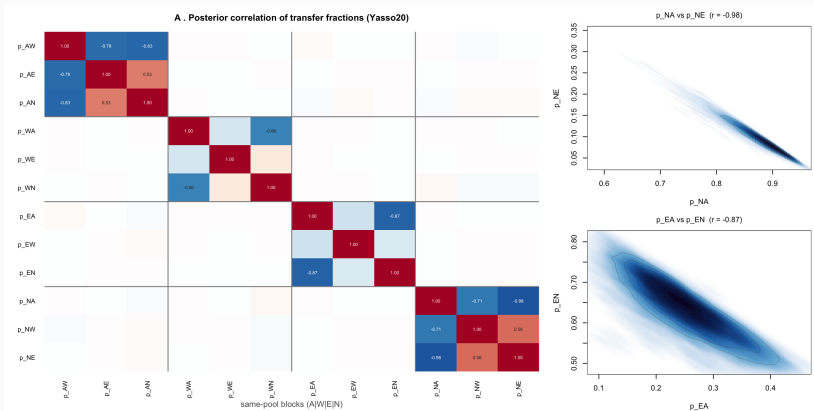
- **The effect of the starting state is not isolated:** the run started in balance with present input has never been done (design ready).
- Sensitivity to the declared error total (0.800): test, or state untested.
- The 1917 stock each model implies: report it, and source a comparison floor.
- Prior-criteria table; the “implications for modelling” subsection; data statement.
- Author order and affiliations.

Backup — the benchmark trio rides the same ridge



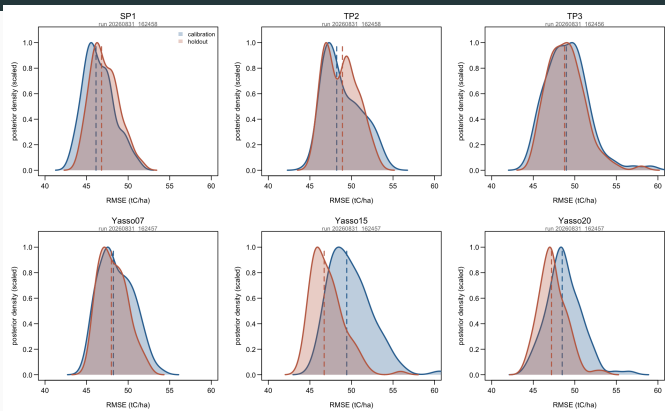
SP1 / TP2 / TP3, same construction as F14. With the climate modifier included: $r = -0.76$ / -0.47 / -0.45 (Yassos -0.59 / -0.53 / -0.56). The trade-off is a property of the data, not of the five-pool structure.

Backup — why the fractions are not identified (and why that is fine)



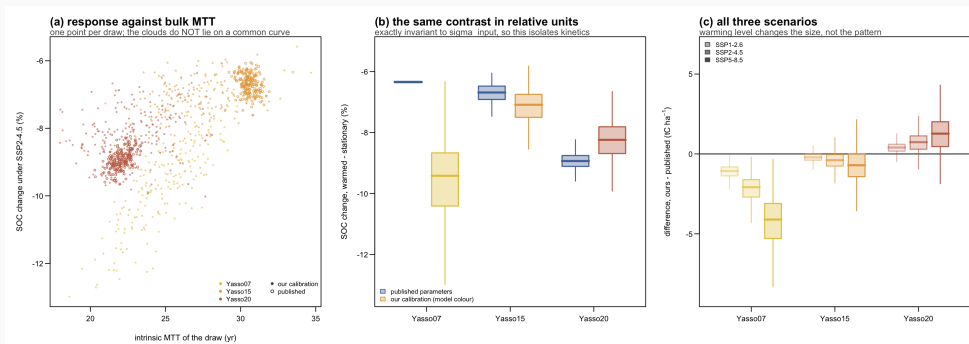
Fractions leaving the same pool are near-perfectly anti-correlated: the data fix the net flux out of a pool, not its split. Prior-pinned, ridge-correlated, < 1 nat — with clean R-hat.

Backup — how tightly is the fit pinned?



RMSE recomputed for every posterior draw. Location = fit; width = how much the posterior disagrees with itself. The cross-run comparison instrument, since log-likelihoods are not comparable across error models.

Backup — why only Yasso07 separates in projection



Yasso07 applies one ξ to every pool: $MTT = MTT_{ref}/\xi$ exactly, so turnover and temperature sensitivity are the same lever. Yasso15/20 carry ξ_{AWE} , ξ_N , ξ_H ; the humus pool that holds the carbon barely moves ($\times 1.04$ – 1.06).

Backup — the likelihood

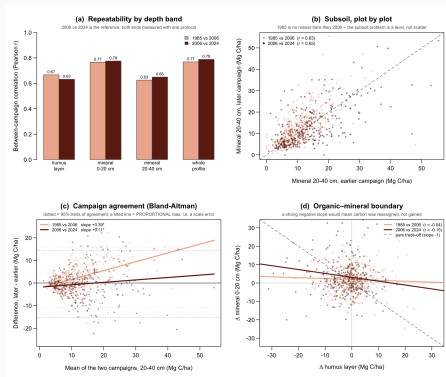
Likelihood (plot i , campaign j , log scale)

$$\log y_{ij} = \log f_{ij}(\theta) + u_{r(i)}^R + u_i^P + u_j^C + e_{ij}$$

term	what it represents	SD
y_{ij}	measured SOC stock	
$f_{ij}(\theta)$	model prediction for parameters θ	
$u_{r(i)}^R$	region: shared by all plots in a latitude band	0.117
u_i^P	plot: shared by all campaigns on a plot (soil, history)	0.396
u_j^C	campaign: shared by all plots of a campaign (protocol)	0.06 / 0.03
e_{ij}	residual: independent, per measurement	0.685
total	declared, split among the four	0.800

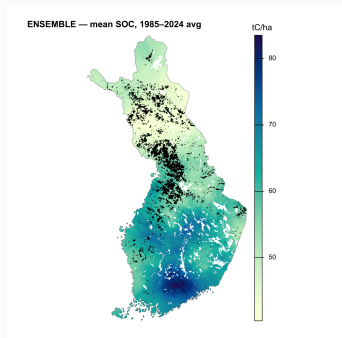
- The offsets are **integrated out**, not estimated: observations that share a plot, a region or a campaign count as less independent information.
- The total is a design value, about twice the spread of the observed stocks around each campaign mean (0.39); chosen high on purpose, so that claims are conservative.

Backup — the observed rate: what it rests on

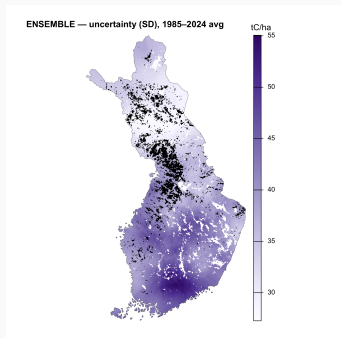


- Basis (decided): balanced set $n = 310$, unweighted, whole profile, true years. Worth $1.8\times$ on the 2006–2024 rate (+0.117 vs +0.209 pairwise).
- 1985 organic layer was OFH only; the missing litter+moss layer (3.4–4.3 Mg C/ha) was 89% of the apparent 1985→2006 organic gain. Each plot's own 2006 LM added.
- “1985” is 1986–1995 (19.5% of plots in 1995): interval to 2024 is 35.1 yr, not 39.
- Full-window rate survives a 9% shared offset; the last-interval rate dissolves at 2%.

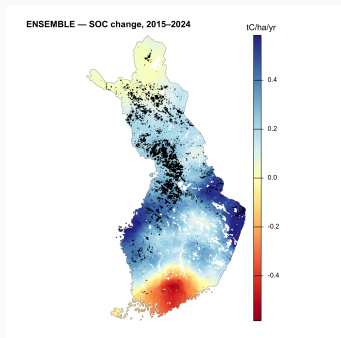
Backup — the deliverable: gridded SOC and its uncertainty



ensemble mean



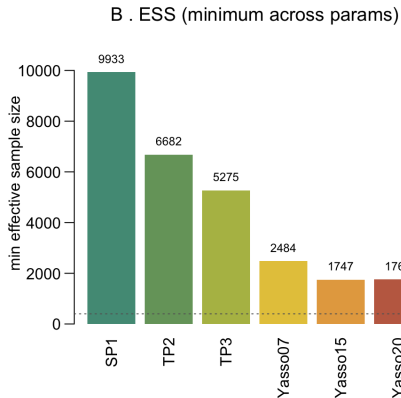
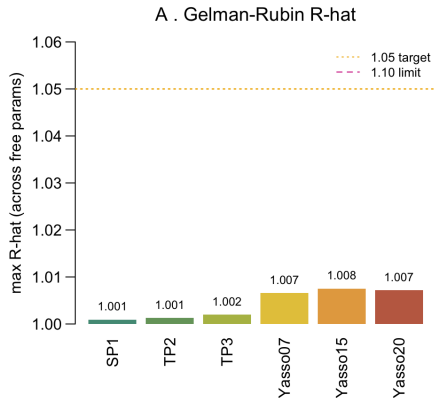
ensemble SD



decadal change

Six-model ensemble, kriged to 2 km, historical 1985–2024 only (no projection). NextGenC / Zenodo deposit, CC-BY-4.0. A bonus product, not a conclusion.

Backup — convergence



All parameters $R\text{-hat} \leq 1.008$, $ESS \geq 1450$, numerically impossible proposals $\sim 0\%$ in all six models (production run 20260831).